

Design-build-fly of a RDC-turbine

NC STATE
UNIVERSITY

Project:

Transform the gaseous rotating detonation combustor (RDC) to a subsonic or supersonic cooled RDC with power extraction. The time of testing will be 5 seconds at 1 lb/sec of oxidizer. Maximum power extraction will be a few kilo-watt at RPMS of 10-50,000 RPM. Size of the combustor is ~ 5 inch. The combustor geometry will be shared with you as well as all the auxiliary systems.

4 intermediate goals

- 1) Integration of a shaft in the upstream plenum
- 2) Design-build-test of a turbine downstream of the combustor to extract power
- 3) Measurements of power **in a reliable way**
- 4) Cooling (gaseous) > you can build off existing cooling schemes or come up with new ones but those then need to be thoroughly vetted

Ultimate goal: proof of concept of a (bladed) turbine that survives for 5 seconds in an RDE environment

Extra information for you:

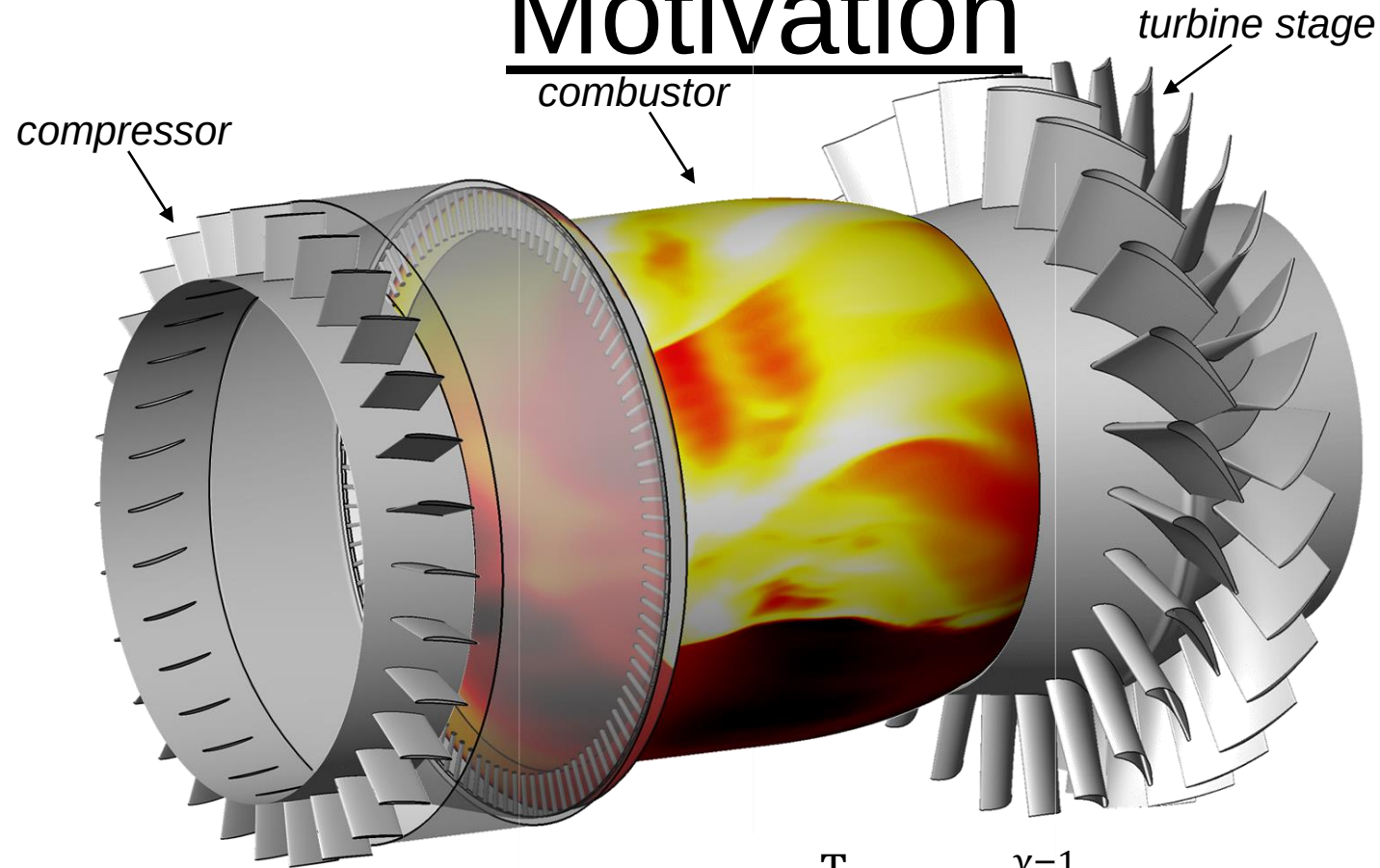
- 1) The combustor itself is already designed and machined and will be tested in November.
- 2) There is a lot of measurement techniques that are already available that can be used by the senior design team
- 3) all auxiliary systems (air , test stand) are already available to use for the students
- 3) Just like with the previous project: safety is again a top priority (health and safety system etc)
- 4) There are publications on cooling available (also from our group)

Budget: ~4000\$

?What is the novelty for this senior design group?

Currently rotating detonation combustors are mainly tested only as a combustor, however the next step is to integrate them with a downstream turbine, and has not yet been shown in lab scale (except for a few trials in the past by the air force and other select labs). Your findings can easily fit within an AIAA conference such as Scitech or ASME Turbo expo.

Motivation



compressor power $\sim \dot{m} c_p \frac{T_{0,\text{inlet}}}{\eta_c} \left(\Pi^{\frac{\gamma-1}{\gamma}} - 1 \right)$

Compressor power reduction in function of
diffuser total pressure drop

	0%	20%	30%
$\Pi \sim 9$	21%	$\sim 11\%$	$\sim 4\%$
$\Pi \sim 7$	24%	$\sim 16\%$	$\sim 8\%$

Types of turbines

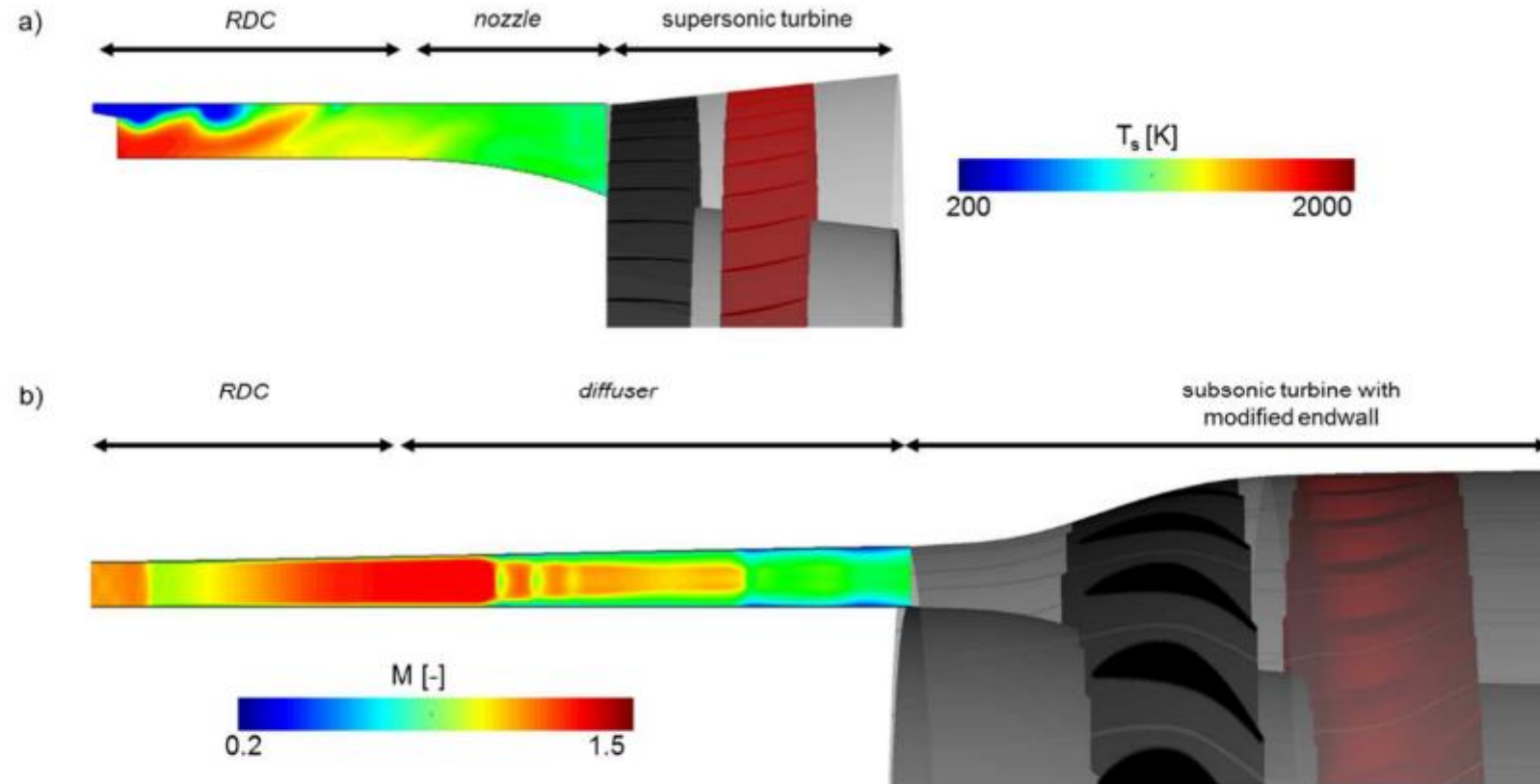
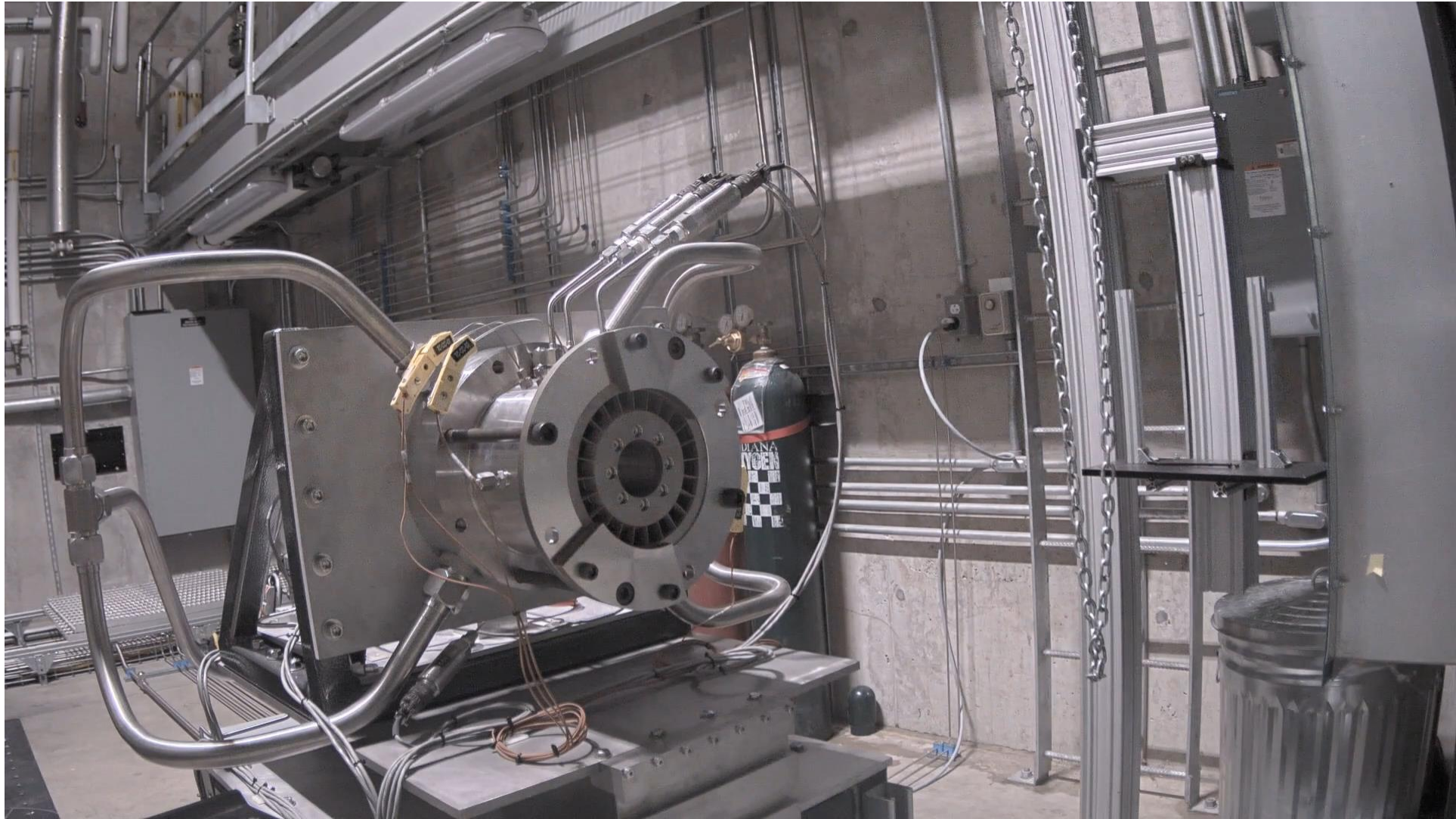


Figure 1. Fluid expansion systems downstream a rotating detonation combustor: a) supersonic turbine + nozzle and b) subsonic turbine + diffuser

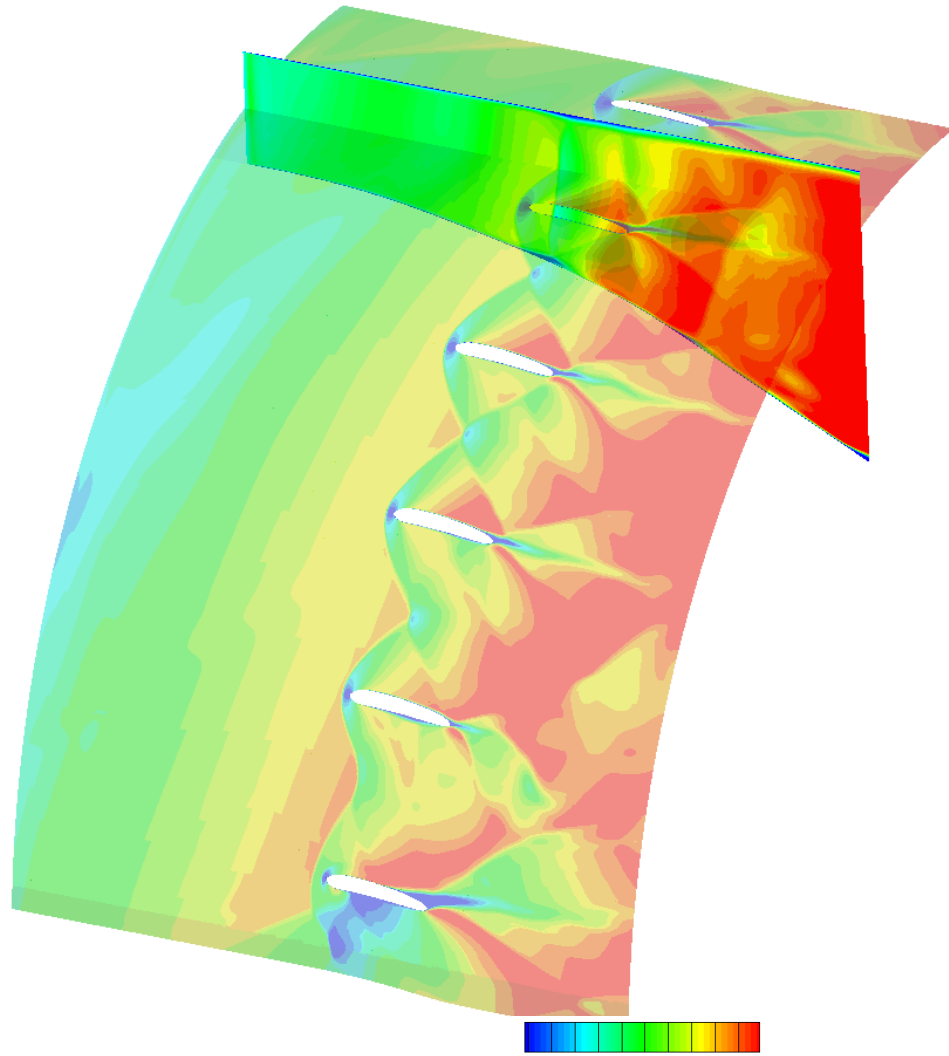
Go for a nozzle and a supersonic nozzle (lower power extraction) or subsonic turbine (with a diffuser)

Integration of supersonic STATOR in THOR

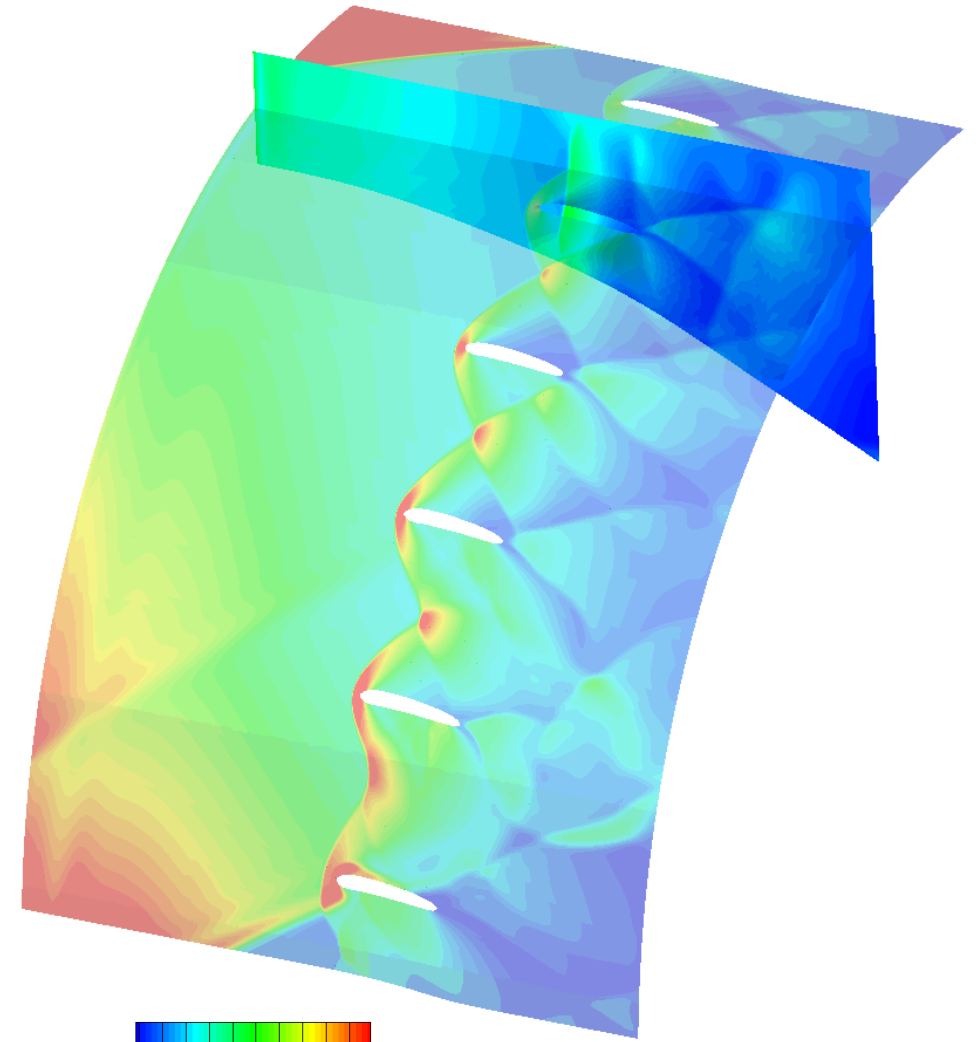


Credits: Braun, Meyer, Athmanathan, Paniagua, Fugger

Nozzle + Stator strategy for RDCs



0.5 M [-] 1.95

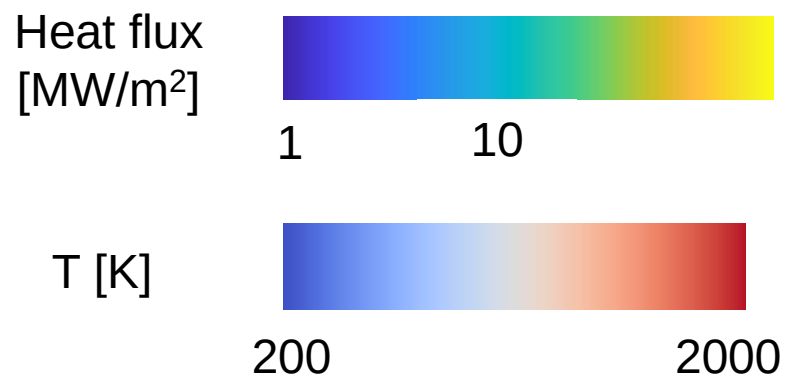


0.5 p [bar] 5

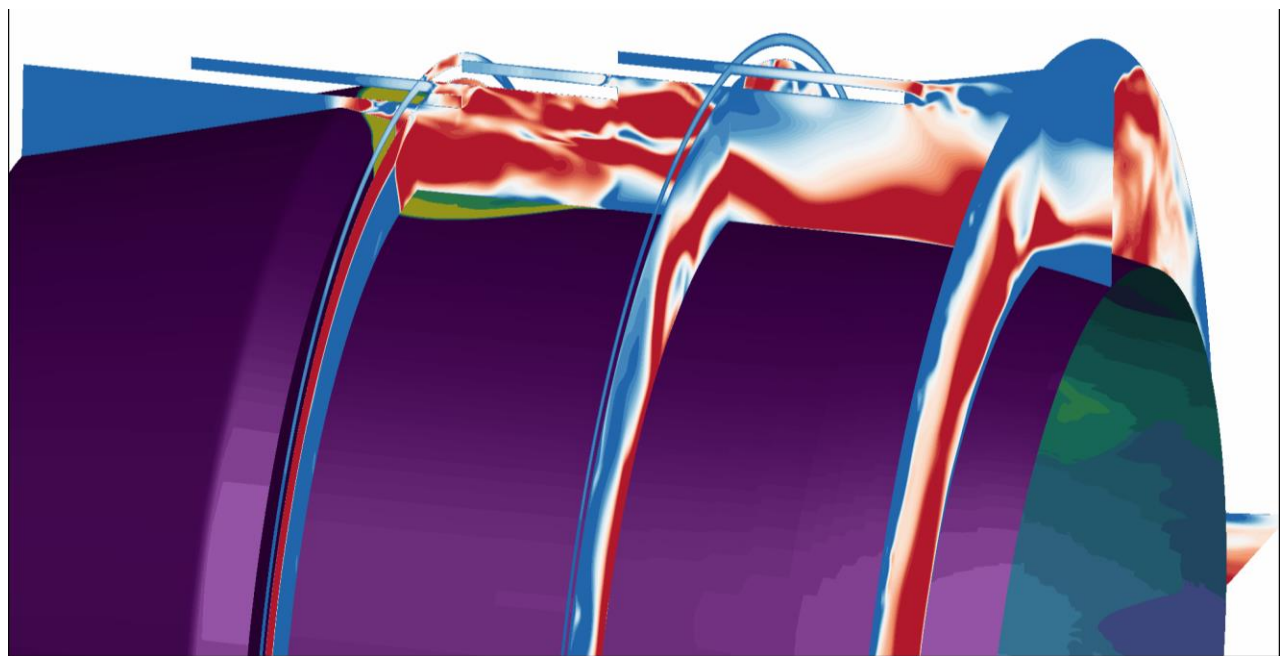
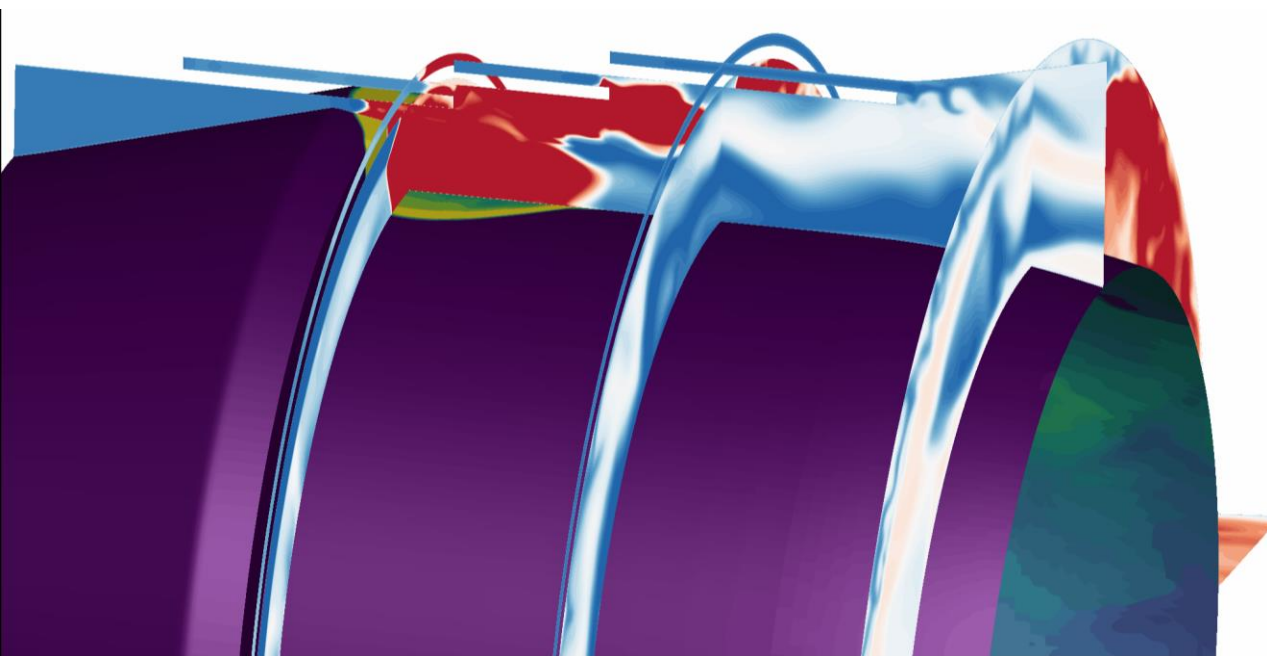
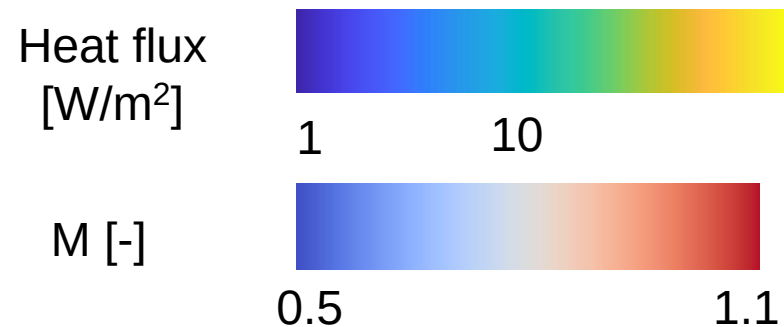
Gaseous cooling

COOLING > Paper Scitech

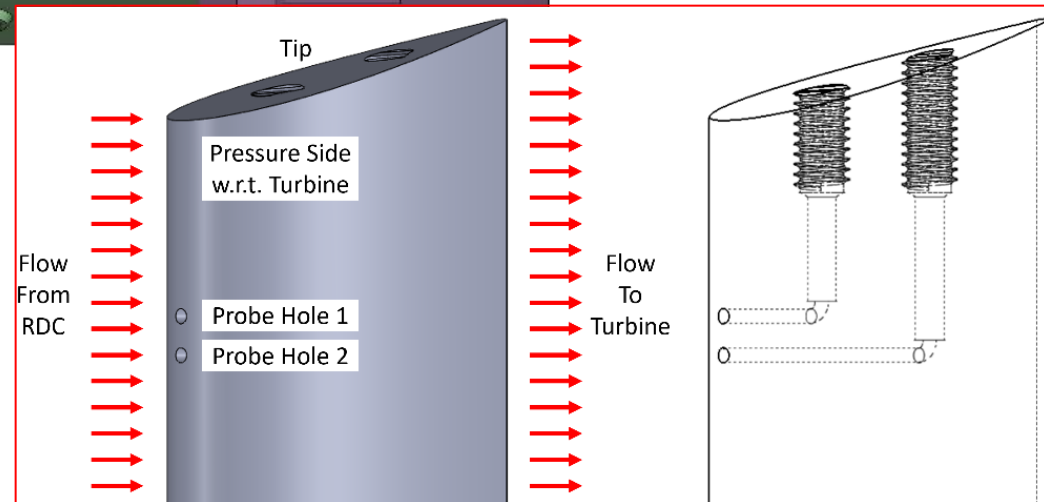
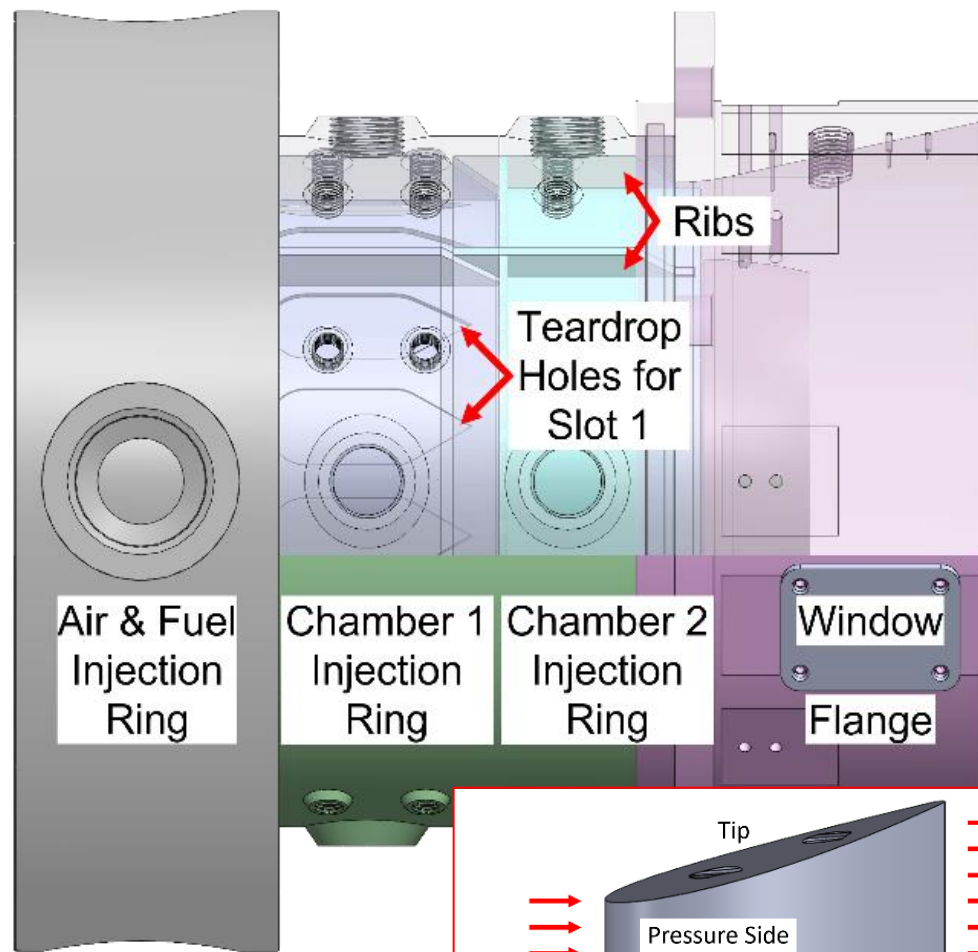
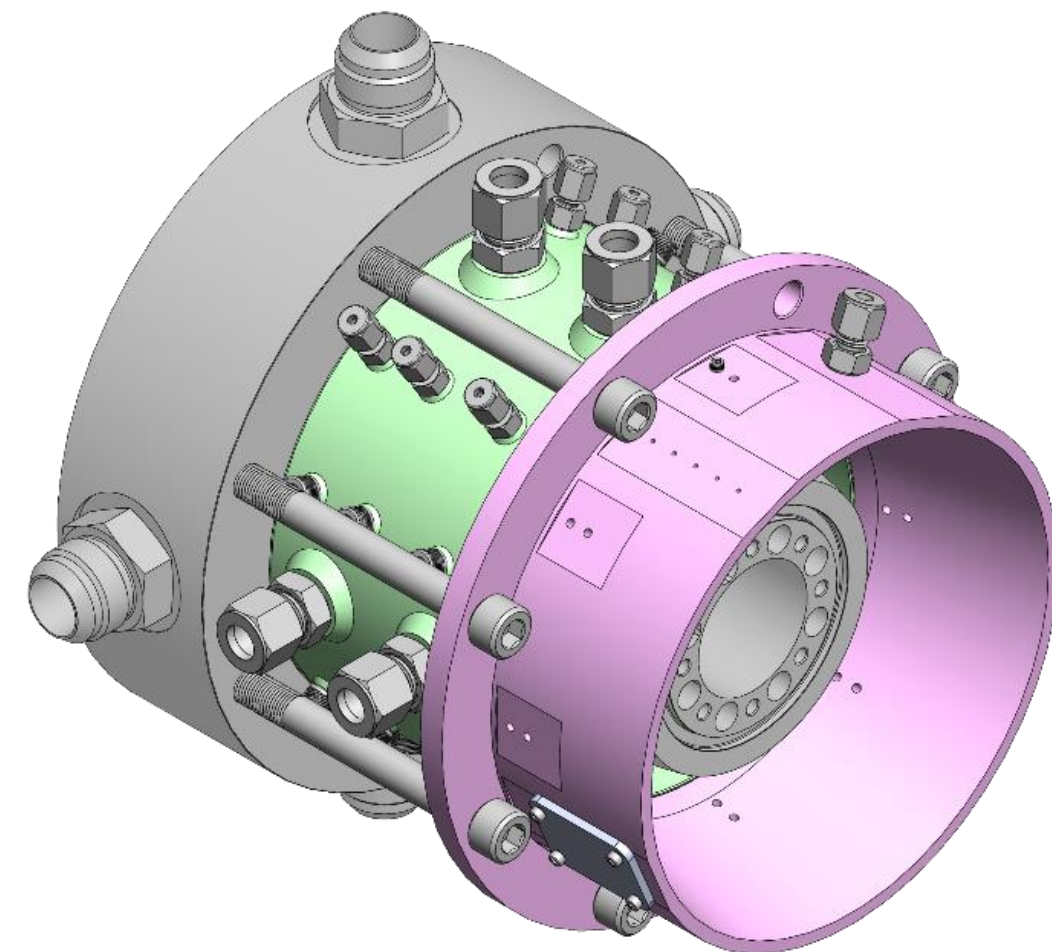
Surface heat flux– Temperature slice



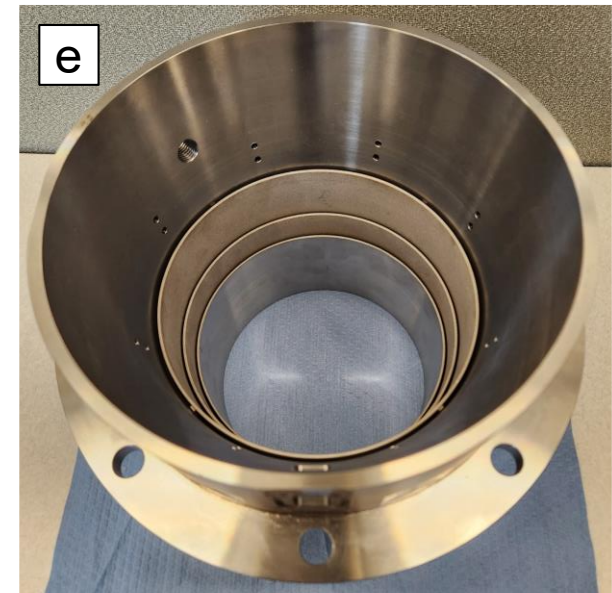
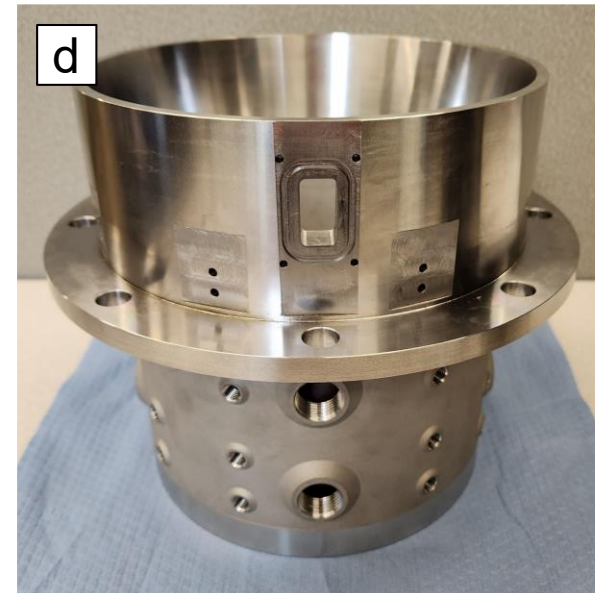
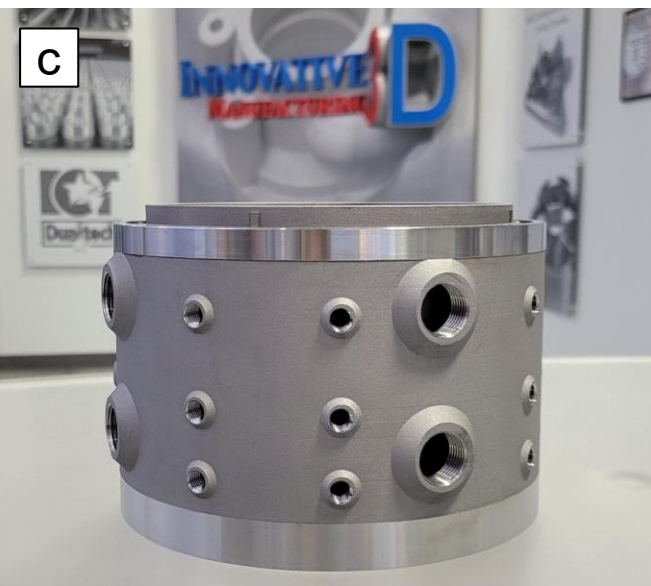
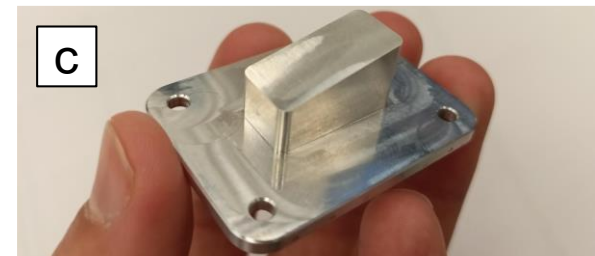
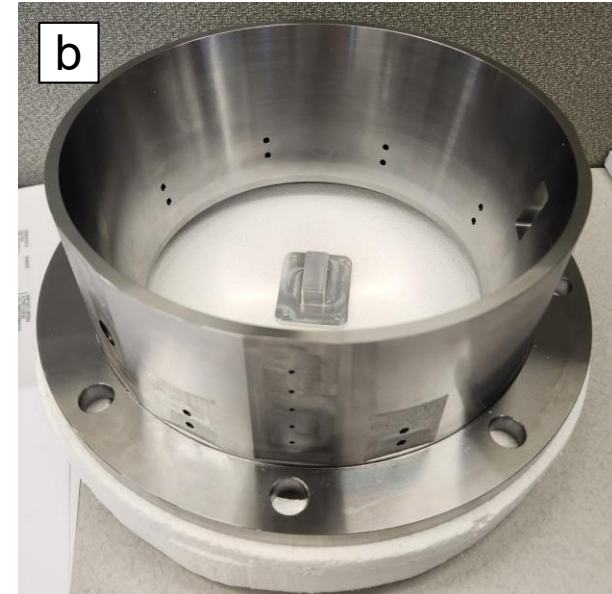
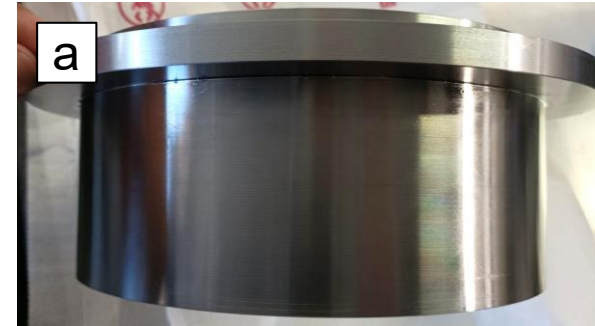
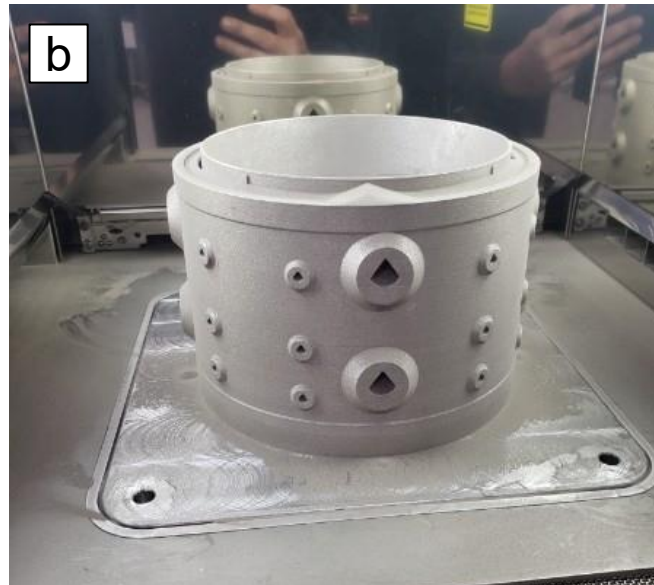
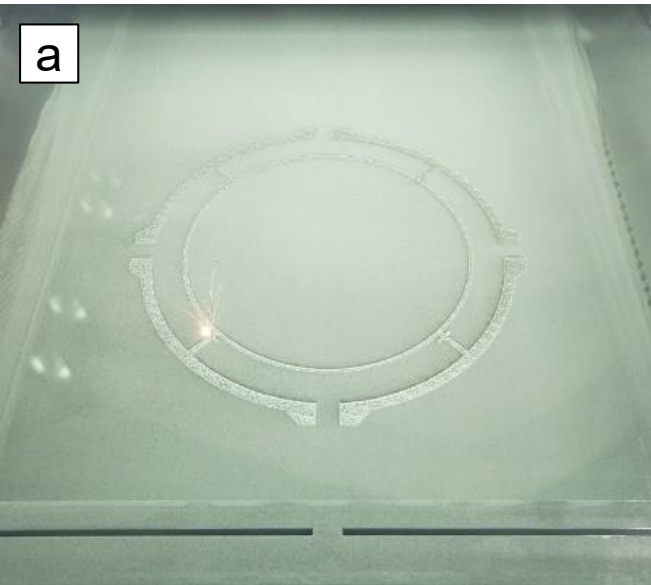
Surface Heat Flux – Mach number slice



CAD design for the 3 slots



Machining



Power absorption: Lynch thesis

